

KAHULUI HARBOR, HI, USA

WMO: 997384

Lat: 20.8950N

Lon: 156.4690W

Elev: 13

StdP: 14.69

Time Zone: -10.00 (W10)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	63.9	65.6	N/A	N/A	N/A	N/A	N/A	N/A	28.1	73.7	26.3	74.0	3.4	160	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	5.6	83.3	N/A	82.3	N/A	81.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	13.8	50

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Extreme Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.9	23.8	21.4	60.3	84.9	1.0	1.8	59.6	86.1	59.1	87.2	58.5	88.2	57.8	89.5		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	74.7	71.8	71.9	71.7	73.1	74.6	75.8	76.9	77.5	77.6	77.0	75.2	73.4		
	DBStd	2.84	2.01	2.18	1.96	1.82	1.64	1.24	1.62	1.60	1.95	1.73	1.96	1.98		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	9025	677	612	674	693	761	773	833	854	828	837	757	726		
	CDD65	3550	212	192	209	243	296	323	368	389	378	372	307	261		
	CDH74	17654	455	403	263	577	1129	1580	2383	2793	2923	2726	1579	843		
	CDH80	635	2	2	1	2	13	23	68	135	211	128	42	8		
Wind	WSAvg	11.6	9.7	10.1	11.9	12.4	11.8	13.6	13.5	12.8	11.7	10.7	11.0	9.9		
Precipitation	PrecAvg	27.6	5.9	3.9	3.8	2.1	1.2	0.4	0.5	0.6	0.5	1.8	3.5	3.3		
	PrecMax	44.7	22.4	13.4	13.0	7.7	4.8	2.8	2.2	1.6	1.7	8.2	14.1	12.4		
	PrecMin	10.8	0.7	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2		
	PrecStd	10.6	5.5	3.7	3.1	2.0	1.4	0.6	0.5	0.4	0.5	2.1	3.7	3.1		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.2	80.2	79.7	80.1	81.5	82.4	83.4	84.3	85.2	83.9	83.0	81.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	78.5	78.8	77.5	78.7	80.1	80.4	81.8	82.4	83.3	82.4	81.2	79.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	77.5	77.6	76.4	77.7	79.0	79.4	80.7	81.5	82.2	81.4	80.1	78.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	76.5	76.5	75.4	76.9	78.1	78.5	80.0	80.6	81.3	80.5	79.2	77.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Mean Daily Temperature Range	5% DB	MDBR	8.6	7.6	6.2	6.4	6.6	5.3	5.1	5.1	5.6	6.7	6.9	8.0		
		MCDBR	10.8	10.0	8.1	8.1	7.9	6.6	6.0	6.2	6.8	7.6	8.4	9.8		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
Clear-Sky Solar Irradiance	taub		0.387	0.395	0.407	0.420	0.410	0.395	0.396	0.396	0.386	0.397	0.397	0.379		
	taud		2.485	2.460	2.434	2.377	2.440	2.482	2.489	2.489	2.516	2.499	2.488	2.532		
	Ebn at Noon		281	285	285	281	281	283	283	284	287	281	275	280		
	Edh at Noon		32	35	37	40	37	35	35	35	34	33	32	30		
All-Sky Solar Radiation	RadAvg		1423	1603	1797	2051	2182	2246	2221	2158	2019	1710	1445	1302		
	RadStd		60	96	167	121	112	58	54	87	56	44	34	111		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+162	+162

Nomenclature: See separate page